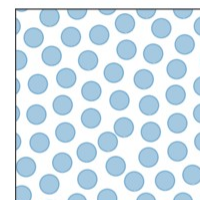


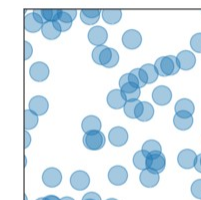
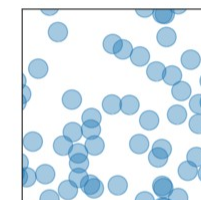
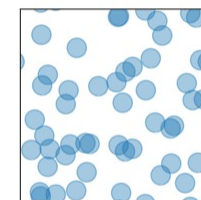
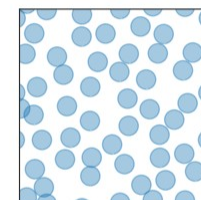
Aula 07 – Integração Termodinâmica

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$$\mathcal{H}_0(\mathbf{q}, \mathbf{p}) = \sum_i \frac{p_i^2}{2m} + \sum_{i \neq j} U_{LJ}(r_{ij})$$



$$\mathcal{H}_t(\mathbf{q}, \mathbf{p}) = \sum_i \frac{p_i^2}{2m} + U_t^\theta(\mathbf{q})$$



t

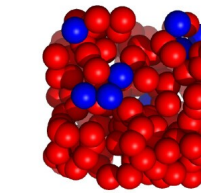
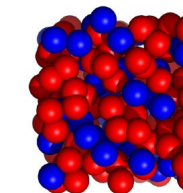
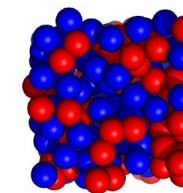
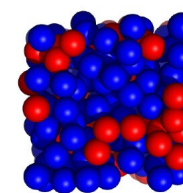
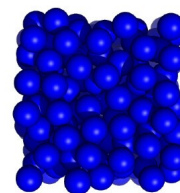
$$\hat{Z}_{LJ}^N = Z_{\text{ideal}}^N \exp \left(-\beta \int_0^1 dt \langle \partial_t U_t^\theta \rangle \right)$$

Thermodynamic integration

$$Z_{\text{ideal}}^N = \frac{(V\Lambda^{-d})^N}{N!}$$

t

$$\mathcal{H}_0(\mathbf{q}, \mathbf{p}) = \sum_i \frac{p_i^2}{2m} + \sum_{i \neq j} U_{LJ}(r_{ij})$$



$$\hat{Z}_{LJ}^N = Z_{\text{ideal}}^N \exp \left(\beta \int_0^1 dt \langle \partial_t U_t^\theta \rangle \right)$$

Thermodynamic integration

$$Z_{\text{ideal}}^N = \frac{(V\Lambda^{-d})^N}{N!}$$

Imagens: Máté, B., Fleuret, F., & Bereau, T. (2024). Neural Thermodynamic Integration: Free Energies from Energy-Based Diffusion Models. The Journal of Physical Chemistry Letters, 15(45), 11395-11404.

Energia Livre e Simulação Molecular

- Hamiltoniano e Energia Total

Energia Térmica

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i} + \sum_{j>i} \sum_{i=1}^N u(\mathbf{q}_{ij})$$

$$\beta = 1/k_B T$$

- Função de Partição Canônica

$$Z = \frac{1}{N! h^{3N}} \int \cdots \int e^{-\beta H(\mathbf{q}, \mathbf{p})} d\mathbf{q}^N d\mathbf{p}^N = \frac{Z_{\text{ideal}}^N}{V^N} \left[\int e^{-\beta \sum_{j>i} \sum_{i=1}^N u(\mathbf{q}_{ij})} d\mathbf{q}^N \right]$$

$$Z_{\text{ideal}}^N = \frac{V^N}{N! h^{3N}} \left[\int e^{-\beta \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i}} d\mathbf{p}^N \right] = \frac{V^N \Lambda^{-3N}}{N!}$$

Energia Livre de Helmholtz (NVT)

$$F = -k_B T \ln Z$$

Integração Termodinâmica (IT)

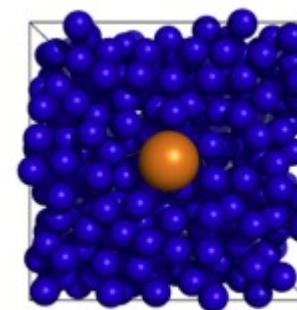
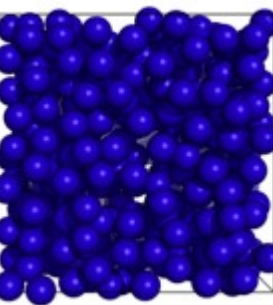
- Generalizando Interação soluto e solvente

$$U_\lambda = U_A + U_B + U_{AB}^\lambda$$

$$U_{AB}^{\lambda=0} = 0 \quad U_{AB}^{\lambda=1} = U_{AB}$$

- Diferença de energia livre

$$\begin{aligned} \Delta F_{0 \rightarrow 1} &= \int_0^1 \frac{\partial F}{\partial \lambda} d\lambda = -k_B T \int_0^1 \frac{\partial \ln Z_\lambda}{\partial \lambda} d\lambda \\ &= \int_0^1 \frac{1}{Z_\lambda} \frac{\partial}{\partial \lambda} \left[\int e^{-\beta U_\lambda(x)} dx \right] d\lambda \\ &= \int_0^1 \frac{1}{Z_\lambda} \left[\int e^{-\beta U_\lambda(x)} \frac{\partial U_\lambda(x)}{\partial \lambda} dx \right] d\lambda \end{aligned}$$



$$U_0 = U_A + U_B$$

$$U_1 = U_A + U_B + U_{AB}$$

Integração Termodinâmica

$$\Delta F_{0 \rightarrow 1} = \int_0^1 \left\langle \frac{\partial U_\lambda(x)}{\partial \lambda} \right\rangle_\lambda d\lambda$$



IT na Dinâmica Molecular

$$\Delta F_{0 \rightarrow 1} = \frac{1}{N_\lambda} \sum_{\lambda_i} \left\langle \frac{\partial U_\lambda(x)}{\partial \lambda} \right\rangle_{\lambda_i}$$

$$\lambda_i \in \{0, 1/N_\lambda, 2/N_\lambda, \dots, (N_\lambda - 1)/N_\lambda\}$$

